Linear Functionals, Dual Spaces, and the Riesz Representation Theorem

From a Data Science Perspective

Biajid Ejaj Chowdhury

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Abstract

Measurement is one of the fundamental aspects underlying mathematics, science, engineering, and data analysis. At its simplest, measurement associates a numerical value with an object or phenomenon. As the objects under consideration become more complex, however, a single real number is often insufficient to describe them. Vectors, matrices, and tensors arise naturally as increasingly rich mathematical representations of data, and correspondingly richer mathematical mechanisms are required to extract meaningful numerical information from them.

This note develops the theory of linear functionals and dual spaces from this measurement-oriented perspective. Particular attention is given to the distinction between vectors and linear functionals, dual bases, changes of basis, covariant and contravariant coordinates, Euclidean spaces, metric coefficients, and the identification of vectors with linear functionals through an inner product. This development culminates in the finite-dimensional Riesz Representation Theorem, which establishes that every linear functional on a finite-dimensional inner-product space can be represented uniquely by an inner product with a vector.

Although matrices and tensor notation naturally appear throughout the discussion, they will be treated primarily as auxiliary computational and representational devices. The central objects of interest will remain vectors and the linear functionals that measure them. The presentation is motivated by applications in data science, where high-dimensional objects must constantly be transformed into meaningful scalar quantities.

1 Introduction

Since the earliest stages of civilization, measurement has played a central role in the development of human knowledge. Lengths were compared, land was surveyed, quantities of food and goods were counted, time was recorded, and astronomical observations were quantified. Although many animals exhibit an ability to distinguish approximate quantities, distances, magnitudes, or durations, human beings developed something substantially more systematic: symbolic numerical systems, standardized units, measuring instruments, and abstract mathematical theories of measurement.

In this sense, the history of civilization may also be viewed partly as a history of increasingly sophisticated methods of assigning numbers to the world around us.

At the most elementary level, measurement produces a real number. If the length of an object is measured, for example, one might obtain, for example, 2.7 meters.

This statement contains several ingredients. There is an object being measured, a quantity of interest, a numerical value, and a chosen unit. Over time, civilizations developed numerical systems, standardized units, and increasingly sophisticated measuring devices so that such numerical descriptions could be made reproducible and useful.

However, as scientific and mathematical problems became more complicated, a single number was often no longer sufficient to describe an object.

For example, the location of a point in three-dimensional space may require three numbers,

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

and a modern data set may represent one observation by hundreds, thousands, or even millions of numerical attributes. Such objects are naturally represented by vectors.

This creates a new measurement problem.

If the object being studied is no longer a scalar but a vector,

$$x \in V,$$

how should one extract a meaningful numerical measurement from it?

A natural answer is to use a function

$$f : V \longrightarrow \mathbb{R}$$

that accepts a vector as input and produces a scalar as output. When this function respects the linear structure of the vector space, it is called a *linear functional*.

Thus, from the perspective adopted in this note, a linear functional may be regarded as a mathematical measuring device:

$$\boxed{\text{vector} \xrightarrow{\text{linear functional}} \text{real number}}.$$

For example, if

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

then

$$f(x) = 2x_1 - 3x_2 + 5x_3$$

is a linear functional. It takes an object containing three numerical components and compresses it into a single scalar quantity.

This viewpoint is particularly natural in data science. A data point is often represented by a vector

$$x \in \mathbb{R}^n,$$

while many operations of practical interest produce scalar outputs. A linear model computes expressions such as

$$w^T x,$$

a projection measures the component of a vector along a particular direction, an optimization method repeatedly evaluates directional quantities, and statistical procedures frequently summarize high-dimensional observations through scalar-valued functions. Linear functionals therefore provide one of the simplest mathematical models of the general operation

$$\text{high-dimensional data} \longrightarrow \text{scalar information.}$$

As mathematical objects became still more complicated, matrices and tensors were introduced to represent relationships involving multiple directions, indices, or modes of variation. Correspondingly, more elaborate notions of duality and multilinear measurement developed. Those subjects lead naturally toward tensor analysis, differential geometry, statistics, machine learning, and modern numerical computation.

The purpose of the present note, however, is deliberately narrower. Our primary focus will remain on vectors and on linear functionals as devices for extracting scalar information from vectors. Matrices will appear frequently because they provide an efficient coordinate representation of linear transformations, changes of basis, and inner products. Some tensor notation will also arise naturally through indexed quantities such as

$$g_{ij}, \qquad g^{ij}, \qquad x^i, \qquad x_i.$$

Nevertheless, these objects will initially be treated as supporting machinery rather than as the principal subject of study.

A central theme of the discussion will be the distinction between a mathematical object and the coordinates used to describe that object. Changing a basis changes the numerical representation of a vector, but it does not change the vector itself. Likewise, the coordinates of a linear functional may change when the basis changes, while the functional itself remains unchanged. Consequently, the scalar measurement produced when a functional acts on a vector must remain invariant:

$$f(x) = a_i x^i = a'_i x'^i.$$

This simple observation leads naturally to the concepts of *contravariant* and *covariant* coordinates.

When the vector space is additionally equipped with an inner product, an important new connection appears. Every vector y defines a linear functional by

$$f_y(x) = \langle x, y \rangle.$$

The natural reverse question is then:

> Given an arbitrary linear functional f, does there exist a vector y such that
>
> $$f(x) = \langle x, y \rangle$$
>
> for every vector x?

In a finite-dimensional inner-product space, the answer is yes, and the representing vector is unique. This is the content of the *Riesz Representation Theorem.*

The theorem provides a remarkable bridge between a vector space and its dual:

$$\boxed{V \longleftrightarrow V^*}$$

through the inner product. In non-orthonormal coordinates, this correspondence is encoded by the metric coefficients

$$g_{ij} = \langle e_i, e_j \rangle,$$

which relate contravariant and covariant coordinates according to

$$x_i = g_{ij} x^j,$$

while the inverse metric

$$[g^{ij}] = [g_{ij}]^{-1}$$

recovers

$$x^i = g^{ij} x_j.$$

Thus, ideas that may initially appear to be abstract notation—dual spaces, upper and lower indices, metric matrices, and the Riesz representation—can all be understood as parts of a single question:

$$\boxed{\text{How can we extract meaningful scalar measurements from vector-valued information?}}$$

The remainder of this note develops that question systematically. We will begin by reviewing the necessary background on vector spaces, linear functionals, and matrix multiplication. We will then introduce dual spaces and dual bases, study their behavior under changes of coordinates, develop covariant and contravariant representations, introduce Euclidean spaces and metric coefficients, and finally derive the finite-dimensional Riesz Representation Theorem. The discussion will conclude with a complete three-dimensional example in which all of these ideas can be observed simultaneously.

2 Background and Prerequisites

Before developing the theory of dual spaces and the Riesz Representation Theorem, we review several concepts that will be used throughout this note. Our purpose here is not to give a complete treatment of linear algebra or vector analysis, but rather to establish the notation and computational framework required for the discussion that follows.

2.1 Vector Spaces

A vector space is a collection of objects, called *vectors*, on which two fundamental operations are defined:

1. vector addition,
$$x + y,$$

2. scalar multiplication,
$$\alpha x.$$

For the purposes of this note, all vector spaces will be taken over the real numbers unless stated otherwise. Thus,

$$\alpha, \beta \in \mathbb{R}.$$

Definition 2.1 *A nonempty set V is called a* real vector space *if, for every $x, y, z \in V$ and every $\alpha, \beta \in \mathbb{R}$, the following properties hold:*

1. **Closure under addition:**
$$x + y \in V.$$

2. **Commutativity of addition:**
$$x + y = y + x.$$

3. **Associativity of addition:**
$$(x + y) + z = x + (y + z).$$

4. **Existence of a zero vector:** there exists $0 \in V$ such that
$$x + 0 = x.$$

5. **Existence of additive inverses:** for every $x \in V$, there exists $-x \in V$ such that
$$x + (-x) = 0.$$

6. **Closure under scalar multiplication:**
$$\alpha x \in V.$$

7. **Distributivity over vector addition:**
$$\alpha(x + y) = \alpha x + \alpha y.$$

8. **Distributivity over scalar addition:**

$$(\alpha + \beta)x = \alpha x + \beta x.$$

9. **Compatibility of scalar multiplication:**

$$\alpha(\beta x) = (\alpha\beta)x.$$

10. **Multiplicative identity:**

$$1x = x.$$

The most familiar example is

$$V = \mathbb{R}^n.$$

However, the elements of a vector space need not themselves be lists of numbers. Polynomials, functions, matrices, signals, and many other mathematical objects may form vector spaces.

This abstraction is important from a data-science perspective. A vector space provides a common mathematical framework in which many apparently different kinds of data can be added, scaled, transformed, and measured.

2.2 Linear Combinations, Span, and Basis

Given vectors

$$e_1, e_2, \ldots, e_n \in V,$$

an expression of the form

$$\alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n$$

is called a *linear combination* of these vectors.

The set of all possible linear combinations of $e_1, \ldots, e_n$ is their span:

$$\operatorname{span}\{e_1, \ldots, e_n\}.$$

A collection

$$e_1, \ldots, e_n$$

is called a *basis* of V if the vectors are linearly independent and span V.

Consequently, every vector $x \in V$ can be written uniquely as

$$\boxed{x = x^1 e_1 + x^2 e_2 + \cdots + x^n e_n.}$$

Equivalently,

$$\boxed{x = \sum_{i=1}^{n} x^i e_i.}$$

The numbers

$$x^1, x^2, \ldots, x^n$$

are the *coordinates* of x relative to the chosen basis.

It is essential to distinguish the vector itself from its coordinates. The vector

$$x$$

is an abstract mathematical object, whereas

$$\begin{bmatrix} x^1 \\ x^2 \\ \vdots \\ x^n \end{bmatrix}$$

is a numerical representation of that vector relative to a particular basis.

If the basis changes, these numbers generally change, even though the underlying vector x remains exactly the same.

This distinction will become one of the central themes of our discussion.

2.3 Linear Functionals

We now introduce the principal measuring device considered in this note.

Definition 2.2 *Let V be a real vector space. A function*

$$f : V \longrightarrow \mathbb{R}$$

is called a linear functional *if, for every $x, y \in V$ and every $\alpha, \beta \in \mathbb{R}$,*

$$\boxed{f(\alpha x + \beta y) = \alpha f(x) + \beta f(y).}$$

Thus, a linear functional accepts a vector and returns a scalar.

From the measurement perspective developed in the introduction, we may visualize this operation as

$$\boxed{V \xrightarrow{\ f\ } \mathbb{R}.}$$

For example, define

$$f : \mathbb{R}^3 \longrightarrow \mathbb{R}$$

by

$$f(x_1, x_2, x_3) = 2x_1 - 3x_2 + 5x_3.$$

Then

$$f\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = 2x_1 - 3x_2 + 5x_3.$$

The three-dimensional vector has therefore been converted into a single real number.

In matrix notation,

$$f(x) = \begin{bmatrix} 2 & -3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Thus,

$$f(x) = a^T x,$$

where

$$a = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}.$$

This simple expression already suggests an intimate relationship between linear functionals, vectors, and matrix multiplication. Later we will see that an inner product makes this relationship considerably deeper.

2.4 Matrix–Vector Multiplication

Because matrices will be used extensively to describe changes of basis and coordinate transformations, it is useful to review matrix multiplication from a structural rather than merely computational point of view.

Let

$$A = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{bmatrix},$$

where a_j denotes the j-th column of A.

For a vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

matrix–vector multiplication may be interpreted as a linear combination of the columns of A:

$$\boxed{Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n.}$$

Equivalently,

$$\boxed{Ax = \sum_{j=1}^{n} x_j a_j.}$$

Componentwise, the i-th entry is

$$\boxed{(Ax)_i = \sum_{j=1}^{n} A_{ij} x_j.}$$

Thus, matrix multiplication naturally contains summation over an index. This observation will later motivate Einstein's summation convention.

2.5 Matrix–Matrix Multiplication: Right Multiplication

A distinction that will become particularly important when studying changes of basis is the difference between multiplying a matrix on the right and multiplying it on the left.

Let

$$E = \begin{bmatrix} | & | & & | \\ e_1 & e_2 & \cdots & e_n \\ | & | & & | \end{bmatrix}.$$

The columns of E are the vectors

$$e_1, \dots, e_n.$$

Let A be an $n \times n$ matrix. Consider first the product

$$EA.$$

If the columns of A are denoted by

$$a_1, \dots, a_n,$$

then

$$\boxed{EA = \begin{bmatrix} Ea_1 & Ea_2 & \cdots & Ea_n \end{bmatrix}.}$$

Suppose

$$a_i = \begin{bmatrix} A_{1i} \\ A_{2i} \\ \vdots \\ A_{ni} \end{bmatrix}.$$

Then

$$Ea_i = A_{1i}e_1 + A_{2i}e_2 + \cdots + A_{ni}e_n.$$

Hence,

$$\boxed{Ea_i = \sum_{j=1}^{n} A_{ji}e_j.}$$

Therefore, if we define

$$E' = EA$$

and write

$$E' = \begin{bmatrix} e'_1 & e'_2 & \cdots & e'_n \end{bmatrix},$$

then

$$\boxed{e'_i = \sum_{j=1}^{n} A_{ji}e_j.}$$

This equation will later become our fundamental formula for a change of basis.

Right multiplication by A therefore forms new columns of E by taking linear combinations of the old columns.

2.6 Matrix–Matrix Multiplication: Left Multiplication

Now consider the reverse product

$$AE.$$

Using the column interpretation,

$$\boxed{AE = \begin{bmatrix} Ae_1 & Ae_2 & \cdots & Ae_n \end{bmatrix}.}$$

Thus, each column of E is acted upon individually by A.

Alternatively, left multiplication can be understood from the row perspective. If

$$A = \begin{bmatrix} - & r_1(A) & - \\ - & r_2(A) & - \\ & \vdots & \\ - & r_n(A) & - \end{bmatrix},$$

then

$$\boxed{AE = \begin{bmatrix} - & r_1(A)E & - \\ - & r_2(A)E & - \\ & \vdots & \\ - & r_n(A)E & - \end{bmatrix}.}$$

Thus, left multiplication naturally forms new rows through combinations determined by the rows of A.

2.7 Comparing EA and AE

The difference between the two products becomes especially transparent in component notation.

For right multiplication,

$$\boxed{(EA)_{ij} = \sum_{k=1}^{n} E_{ik} A_{kj}.}$$

For left multiplication,

$$\boxed{(AE)_{ij} = \sum_{k=1}^{n} A_{ik} E_{kj}.}$$

Thus,

$$EA \qquad \text{and} \qquad AE$$

are generally different matrices.

The order of multiplication matters:

$$\boxed{EA \neq AE \quad \text{in general.}}$$

For our later discussion of basis changes, EA will be particularly important because the basis vectors are stored as the columns of E. Consequently,

$$E' = EA$$

means precisely that the new basis vectors are linear combinations of the old basis vectors:

$$\boxed{e'_i = \sum_j A_{ji} e_j.}$$

2.8 Index Notation

Expressions involving vectors, matrices, and changes of coordinates often contain many summations. Index notation provides a compact way of expressing these relationships.

For example,

$$x = \sum_{i=1}^{n} x^i e_i$$

expresses a vector as a linear combination of basis vectors.

Similarly,

$$e'_i = \sum_{j=1}^{n} A_{ji} e_j$$

describes a change of basis.

The letters $i, j, k, \ldots$ used in such expressions are called *indices*. An index that is summed over is often called a *dummy index*. Its particular letter is irrelevant. Thus,

$$\sum_{j=1}^{n} A_{ji} e_j$$

and

$$\sum_{k=1}^{n} A_{ki} e_k$$

represent exactly the same quantity.

By contrast, an index that remains after the summation is called a *free index*. For example, in

$$e'_i = \sum_{j} A_{ji} e_j,$$

j is a dummy index while i is a free index.

2.9 Einstein Summation Convention

In many areas of mathematics, physics, statistics, and tensor analysis, explicit summation symbols are suppressed using the *Einstein summation convention*.

Whenever an index appears once as an upper index and once as a lower index, summation over that index is understood.

Thus,

$$\sum_{i=1}^{n} x^i e_i$$

may be written simply as

$$\boxed{x = x^i e_i.}$$

Likewise,

$$\sum_{i=1}^{n} a_i x^i$$

becomes

$$\boxed{a_i x^i.}$$

The repeated index i is implicitly summed over.

For example,

$$a_i x^i$$

means

$$a_1 x^1 + a_2 x^2 + \cdots + a_n x^n.$$

Similarly,

$$g_{ij} x^j$$

means

$$\sum_{j=1}^{n} g_{ij} x^j.$$

Here j is summed over, while i remains free.

It is useful to distinguish these two types of indices:

- A *dummy index* is summed over and disappears from the final result.
- A *free index* is not summed over and identifies the component of the resulting object.

For example,

$$a_i = g_{ij}x^j$$

contains the repeated dummy index j, while i is the free index.

The position of an index—upper or lower—will eventually acquire a precise mathematical meaning. Upper indices will be associated with *contravariant coordinates*, while lower indices will be associated with *covariant coordinates*. At present, however, we will regard the notation primarily as a convenient bookkeeping system. Its deeper meaning will emerge naturally from the study of changes of basis and dual spaces.

2.10 The Kronecker Delta

Before proceeding to more compact forms of index notation, it is useful to introduce a simple but extremely important object called the *Kronecker delta*.

The Kronecker delta is defined by

$$\boxed{\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}}$$

Thus, the Kronecker delta behaves like an indicator that determines whether two indices are equal.

For example,

$$\delta_{11} = 1, \qquad \delta_{22} = 1, \qquad \delta_{12} = 0, \qquad \delta_{23} = 0.$$

In three dimensions, the collection of Kronecker delta values may be represented by the identity matrix:

$$[\delta_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus,

$$\boxed{[\delta_{ij}] = I.}$$

The Selection Property

The most important property of the Kronecker delta is its ability to *select* a particular term from a summation.

Consider

$$\sum_{j=1}^{n} \delta_{ij} x_j.$$

For a fixed value of i, every term vanishes except the term for which $j = i$. Therefore,

$$\boxed{\sum_{j=1}^{n} \delta_{ij} x_j = x_i.}$$

For example, if $i = 2$ in three dimensions,

$$\begin{aligned} \sum_{j=1}^{3} \delta_{2j} x_j &= \delta_{21} x_1 + \delta_{22} x_2 + \delta_{23} x_3 \\ &= 0x_1 + 1x_2 + 0x_3 \\ &= x_2. \end{aligned}$$

Hence, the Kronecker delta acts as an *index selector*.

Similarly,

$$\boxed{\delta_{ij} x_j = x_i}$$

when Einstein summation notation is being used.

Kronecker Delta with Upper and Lower Indices

Later in this note, we will frequently write the Kronecker delta as

$$\boxed{\delta^i_j.}$$

It has the same basic definition:

$$\delta^i_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Thus,

$$\delta^1_1 = 1, \qquad \delta^1_2 = 0, \qquad \delta^2_1 = 0, \qquad \delta^2_2 = 1.$$

The mixed-index notation becomes particularly convenient when using Einstein summation convention, because an upper index and a lower index naturally form a summation pair.

For example,

$$\boxed{\delta^i_j x^j = x^i.}$$

Again, the Kronecker delta simply selects the component whose index matches the free index i.

Kronecker Delta as the Identity Operator

The Kronecker delta may also be understood as the component representation of the identity transformation.

Let

$$I : V \to V$$

denote the identity map,

$$I(x) = x.$$

In coordinates,

$$(Ix)^i = \delta^i_j x^j.$$

But since

$$\delta^i_j x^j = x^i,$$

we obtain

$$(Ix)^i = x^i.$$

Thus,

$$\boxed{\delta^i_j \quad \text{plays the role of the identity in index notation.}}$$

This interpretation will become especially useful later when we encounter relations such as

$$g^{ij} g_{jk} = \delta^i_k,$$

which is simply the indexed version of the matrix identity

$$G^{-1}G = I.$$

Why the Kronecker Delta Matters for Dual Spaces

The Kronecker delta will play a central role when we construct the dual basis.

If

$$e_1, \ldots, e_n$$

is a basis of V, its dual basis

$$f^1, \ldots, f^n$$

will be defined by the condition

$$\boxed{f^i(e_j) = \delta^i_j.}$$

This means that f^i returns 1 when applied to its corresponding basis vector e_i, and returns 0 when applied to every other basis vector.

Consequently, when

$$x = x^j e_j,$$

we obtain

$$\begin{aligned} f^i(x) &= f^i(x^j e_j) \\ &= x^j f^i(e_j) \\ &= x^j \delta^i_j \\ &= x^i. \end{aligned}$$

Thus, the Kronecker delta is precisely what allows a dual basis functional to isolate one coordinate of a vector.

From the measurement perspective of this note, it therefore acts as a perfect coordinate selector:

$$\boxed{\delta^i_j \quad \Longrightarrow \quad \text{select the } i\text{-th component.}}$$

2.11 Summary of the Required Background

The ideas introduced in this section provide the algebraic and computational language for everything that follows. We began with vector spaces, bases, and coordinate representations, and then introduced linear functionals as scalar-valued linear measurements of vectors.

In particular, we will repeatedly use the coordinate expansion

$$x = x^i e_i,$$

the basis-change relation

$$E' = EA,$$

or equivalently,

$$e'_i = A_{ji} e_j,$$

together with the matrix multiplication identities

$$(EA)_{ij} = E_{ik} A_{kj},$$

and

$$(AE)_{ij} = A_{ik}E_{kj}.$$

We also introduced index notation and the Einstein summation convention, according to which a repeated upper–lower index pair is implicitly summed. Thus, for example,

$$x = x^i e_i$$

is shorthand for

$$x = \sum_{i=1}^{n} x^i e_i.$$

The Kronecker delta,

$$\delta^i_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

provides a compact representation of the identity operation in index notation. Its fundamental selection property,

$$\delta^i_j x^j = x^i,$$

allows a particular component to be isolated from a summation. This property will become especially important when we construct the dual basis, whose defining relation will be

$$f^i(e_j) = \delta^i_j.$$

Thus, the principal pieces of notation that we carry forward are

$$\boxed{x = x^i e_i, \qquad e'_i = A_{ji} e_j, \qquad \delta^i_j x^j = x^i.}$$

These ideas allow us to distinguish carefully between a mathematical object and its coordinate representation, while providing an efficient language for describing how those representations change.

We are now prepared to introduce the collection of all linear measuring devices acting on a vector space: the *dual space*.

3 Linear Functionals and the Dual Space

We now move from the vectors being measured to the mathematical objects that perform the measurement. As introduced earlier, a linear functional is a linear map

$$f : V \longrightarrow \mathbb{R}.$$

Thus, while a vector belongs to the vector space V, a linear functional acts on vectors in V and produces real numbers.

This distinction is fundamental:

$$\boxed{x \in V \qquad \text{whereas} \qquad f : V \to \mathbb{R}.}$$

A vector and a linear functional are therefore different kinds of mathematical objects. Nevertheless, as we shall see, the collection of all linear functionals on V possesses a vector-space structure of its own.

3.1 A Linear Functional Is Determined by Its Values on a Basis

Let

$$e_1, \ldots, e_n$$

be a basis of an n-dimensional vector space V. Every vector $x \in V$ has a unique expansion

$$x = x^i e_i.$$

Let f be a linear functional on V. Applying f gives

$$f(x) = f(x^i e_i).$$

By linearity,

$$f(x) = x^i f(e_i).$$

Define

$$\boxed{a_i = f(e_i).}$$

Then

$$\boxed{f(x) = a_i x^i.}$$

Written with the summation displayed explicitly,

$$f(x) = \sum_{i=1}^{n} a_i x^i.$$

Therefore, once the n numbers

$$f(e_1), \ldots, f(e_n)$$

are known, the value of $f(x)$ can be determined for every vector $x \in V$.

Consequently,

$$\boxed{\text{a linear functional is completely determined by its values on a basis.}}$$

This observation is important from a measurement perspective. A linear measuring device need not be tested on every possible vector. It is enough to know how it measures the basis directions; linearity then determines its response to every possible linear combination of those directions.

3.2 Example: Reconstructing a Functional from Basis Values

Consider $V = \mathbb{R}^3$ with the standard basis

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Suppose a linear functional f satisfies

$$f(e_1) = 2, \qquad f(e_2) = -3, \qquad f(e_3) = 5.$$

An arbitrary vector is

$$x = x^1 e_1 + x^2 e_2 + x^3 e_3.$$

Therefore,

$$\begin{aligned} f(x) &= x^1 f(e_1) + x^2 f(e_2) + x^3 f(e_3) \\ &= 2x^1 - 3x^2 + 5x^3. \end{aligned}$$

Thus the three measurements

$$2, \qquad -3, \qquad 5$$

completely determine the functional.

3.3 The Dual Space

We now collect all possible linear functionals on V.

Definition 3.1 *The set of all linear functionals*

$$f : V \longrightarrow \mathbb{R}$$

is called the dual space *of V, and is denoted by*

$$\boxed{V^*.}$$

Thus,

$$V^* = \{f : V \to \mathbb{R} \mid f \text{ is linear}\}.$$

The original space V contains vectors:

$$x, y, z, \ldots$$

whereas the dual space V^* contains linear functionals:

$$f, g, h, \ldots$$

Schematically,

V	V^*
vectors	linear functionals
x	f
y	g
z	h

A useful conceptual distinction is therefore

$$\boxed{\text{data objects live in } V, \qquad \text{linear measuring devices live in } V^*.}$$

3.4 Why V^* Is Itself a Vector Space

The term *dual space* is justified because the collection of linear functionals itself satisfies the axioms of a vector space.

Let

$$f, g \in V^*$$

and let

$$\alpha \in \mathbb{R}.$$

Define addition of functionals pointwise by

$$\boxed{(f+g)(x) = f(x) + g(x),}$$

and scalar multiplication by

$$\boxed{(\alpha f)(x) = \alpha f(x).}$$

We should verify that these new objects remain linear functionals.

For $f + g$,

$$\begin{aligned}(f+g)(\lambda x + \mu y) &= f(\lambda x + \mu y) + g(\lambda x + \mu y) \\ &= \lambda f(x) + \mu f(y) + \lambda g(x) + \mu g(y) \\ &= \lambda (f+g)(x) + \mu (f+g)(y).\end{aligned}$$

Thus,

$$f + g \in V^*.$$

Similarly,

$$\begin{aligned}(\alpha f)(\lambda x + \mu y) &= \alpha f(\lambda x + \mu y) \\ &= \alpha \lambda f(x) + \alpha \mu f(y) \\ &= \lambda (\alpha f)(x) + \mu (\alpha f)(y).\end{aligned}$$

Hence,

$$\alpha f \in V^*.$$

The remaining vector-space properties follow from the corresponding properties of real numbers. Therefore V^* is itself a vector space.

3.5 Dimension of the Dual Space

Suppose

$$\dim V = n.$$

Since every linear functional is completely determined by its n values

$$f(e_1), \ldots, f(e_n),$$

we expect a functional to require n independent numerical parameters.

Indeed,

$$\boxed{\dim V^* = \dim V = n.}$$

Thus V and V^* have the same dimension.

However, an important consideration is necessary:

$\boxed{V \text{ and } V^* \text{ having the same dimension does not mean that vectors and functionals are the same objects.}}$

They are different vector spaces containing different types of objects.

Later, after introducing an inner product, we will discover a natural correspondence between them. This distinction is precisely what makes the Riesz Representation Theorem significant.

3.6 The Dual Basis

Let

$$e_1, \dots, e_n$$

be a basis of V.

We now construct a corresponding basis of V^*.

Define linear functionals

$$f^1, \dots, f^n$$

by requiring

$$\boxed{f^i(e_j) = \delta^i_j,}$$

where the Kronecker delta is defined by

$$\delta^i_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Thus,

$$
\begin{aligned}
f^1(e_1) = 1, \quad f^1(e_2) = 0, \quad \cdots f^1(e_n) &= 0, \\
f^2(e_1) = 0, \quad f^2(e_2) = 1, \quad \cdots f^2(e_n) &= 0,
\end{aligned}
$$

and so forth.

The collection

$$\boxed{f^1, \ldots, f^n}$$

is called the *dual basis* corresponding to

$$e_1, \ldots, e_n.$$

3.7 The Dual Basis as a Coordinate Measuring System

The meaning of the dual basis becomes particularly transparent when it acts on an arbitrary vector

$$x = x^j e_j.$$

Apply f^i:

$$f^i(x) = f^i(x^j e_j).$$

By linearity,

$$f^i(x) = x^j f^i(e_j).$$

Using

$$f^i(e_j) = \delta^i_j,$$

we obtain

$$f^i(x) = x^j \delta^i_j.$$

The Kronecker delta eliminates every term except the one for which $j = i$. Therefore,

$$\boxed{f^i(x) = x^i.}$$

This equation gives the dual basis a direct measurement interpretation:

$$\boxed{f^i \text{ measures the } i\text{-th coordinate of a vector.}}$$

For example, in $\mathbb{R}^3$,

$$x = x^1 e_1 + x^2 e_2 + x^3 e_3,$$

and the dual basis functionals return

$$f^1(x) = x^1, \qquad f^2(x) = x^2, \qquad f^3(x) = x^3.$$

Thus the dual basis may be viewed as a collection of coordinate sensors:

$$\boxed{\begin{array}{ccc} f^1(x) & = & x^1, \\ f^2(x) & = & x^2, \\ \vdots & & \vdots \\ f^n(x) & = & x^n. \end{array}}$$

3.8 Expanding a Functional in the Dual Basis

Since

$$f^1, \dots, f^n$$

form a basis of V^*, every functional $f \in V^*$ can be expressed as

$$f = a_i f^i.$$

To determine the coefficients a_i, apply both sides to a basis vector e_j:

$$f(e_j) = a_i f^i(e_j).$$

Since

$$f^i(e_j) = \delta^i_j,$$

we obtain

$$f(e_j) = a_i \delta^i_j = a_j.$$

Therefore,

$$\boxed{a_j = f(e_j).}$$

Hence,

$$\boxed{f = f(e_i) f^i.}$$

The coefficients of a functional in the dual basis are precisely its values on the corresponding basis vectors.

3.9 The Natural Pairing Between V^* and V

Let

$$x = x^i e_i$$

and

$$f = a_i f^i.$$

Then

$$\begin{aligned} f(x) &= (a_i f^i)(x^j e_j) \\ &= a_i x^j f^i(e_j) \\ &= a_i x^j \delta^i_j. \end{aligned}$$

Therefore,

$$\boxed{f(x) = a_i x^i.}$$

Explicitly,

$$f(x) = a_1 x^1 + \cdots + a_n x^n.$$

This scalar is called the *natural pairing* between the functional $f \in V^*$ and the vector $x \in V$.

We may represent the operation schematically as

$$\boxed{V^* \times V \longrightarrow \mathbb{R}, \qquad (f, x) \longmapsto f(x).}$$

Notice the structure of the coordinate expression:

$$\boxed{f(x) = a_i x^i.}$$

One index appears below and the other above. At this stage this may appear to be merely notation, but it reflects a fundamental difference between the way the coordinates of vectors and the coordinates of functionals respond to a change of basis.

3.10 A Data-Science Interpretation

The dual-space viewpoint appears naturally in data analysis.

Suppose one observation is represented by a feature vector

$$x = \begin{bmatrix} x^1 \\ x^2 \\ \vdots \\ x^n \end{bmatrix}.$$

A linear scoring rule may assign the scalar

$$s = a_i x^i.$$

The vector x represents the data object, while the coefficients a_i describe a linear measurement applied to that object.

Examples of this basic structure include linear regression scores, classification scores before a nonlinear activation, linear projections, portfolio exposures, and directional measurements.

The abstract distinction

$$x \in V, \qquad f \in V^*$$

therefore corresponds to a very practical distinction:

$$\boxed{\text{data} \qquad \text{versus} \qquad \text{linear measurement of the data.}}$$

This distinction becomes particularly important when the coordinate system used to describe the data changes. The vector and the functional themselves must remain unchanged, while their numerical representations adjust in different ways.

Understanding precisely how those representations change leads to the concepts of *contravariance* and *covariance*, which we develop next.

4 Change of Basis: Contravariant and Covariant Coordinates

The distinction between a mathematical object and its coordinate representation becomes particularly important when the basis of a vector space is changed.

A vector does not change merely because we choose a different basis with which to describe it. Likewise, a linear functional does not change when the coordinate system changes. What changes are the numerical coefficients used to represent these objects.

This observation leads to two different transformation laws:

$$\boxed{\text{vector coordinates transform contravariantly,}}$$

whereas

$$\boxed{\text{functional coordinates transform covariantly.}}$$

The purpose of this section is to derive these transformation laws and to understand why they must differ.

4.1 Changing the Basis

Let

$$e_1, \dots, e_n$$

be a basis of an n-dimensional vector space V, and collect the basis vectors as the columns of the matrix

$$E = \begin{bmatrix} | & | & & | \\ e_1 & e_2 & \cdots & e_n \\ | & | & & | \end{bmatrix}.$$

Suppose we introduce another basis

$$e'_1, \dots, e'_n,$$

and collect these vectors in

$$E' = \begin{bmatrix} | & | & & | \\ e'_1 & e'_2 & \cdots & e'_n \\ | & | & & | \end{bmatrix}.$$

Every new basis vector is a linear combination of the old basis vectors. Therefore, there exists an invertible matrix A such that

$$\boxed{E' = EA.}$$

Why must A be invertible? Because both E and E' represent bases. The new vectors must remain linearly independent, and therefore the transformation from one basis to the other cannot destroy any direction.

If

$$A = [A_{ji}],$$

then the i-th column of $E' = EA$ gives

$$\boxed{e'_i = A_{ji} e_j.}$$

Written with the summation displayed,

$$e'_i = \sum_{j=1}^{n} A_{ji} e_j.$$

Thus, the i-th column of A contains the coordinates of the new basis vector e'_i relative to the old basis.

4.2 The Vector Does Not Change

Now take a vector

$$x \in V.$$

Relative to the old basis,

$$x = x^i e_i.$$

Relative to the new basis,

$$x = x'^i e'_i.$$

These are not two different vectors. They are two coordinate descriptions of the same vector:

$$\boxed{x = x^i e_i = x'^i e'_i.}$$

In matrix notation, let

$$[x]_E = \begin{bmatrix} x^1 \\ \vdots \\ x^n \end{bmatrix}$$

and

$$[x]_{E'} = \begin{bmatrix} x'^1 \\ \vdots \\ x'^n \end{bmatrix}.$$

Then

$$x = E[x]_E$$

and

$$x = E'[x]_{E'}.$$

Since

$$E' = EA,$$

we obtain

$$E[x]_E = EA[x]_{E'}.$$

Because the columns of E are linearly independent,

$$[x]_E = A[x]_{E'}.$$

Therefore,

$$\boxed{[x]_{E'} = A^{-1}[x]_E.}$$

Thus, when the basis transforms according to

$$\boxed{E' = EA,}$$

the coordinates of a vector transform according to

$$\boxed{x' = A^{-1}x.}$$

Here x and x' denote the coordinate columns in the old and new bases, respectively.

4.3 Why Vector Coordinates Are Called Contravariant

The basis transformation uses A:

$$E' = EA,$$

while the vector coordinates use the inverse transformation:

$$x' = A^{-1}x.$$

Thus, the coordinates move in the inverse sense to the basis change:

$$\boxed{\begin{array}{ccc} E & \longrightarrow & E' = EA, \\ x & \longrightarrow & x' = A^{-1}x. \end{array}}$$

For this reason, the coordinates

$$x^1, \ldots, x^n$$

of a vector are called *contravariant coordinates.*

The prefix "contra" reflects this inverse behavior.

This is also the reason that vector coordinates are conventionally written with upper indices:

$$\boxed{x = x^i e_i.}$$

4.4 A Simple Scaling Example

The contravariant behavior becomes intuitive in one dimension.

Suppose the original basis consists of a single vector e_1, and let

$$x = 6e_1.$$

Now choose a new basis vector

$$e_1' = 2e_1.$$

The basis vector has become twice as large.

To describe the same vector x, we must now write

$$x = 3e_1'.$$

Thus,

$$x^1 = 6, \qquad x'^1 = 3.$$

The basis doubled while the coordinate was divided by two:

$$\boxed{e'_1 = 2e_1 \qquad \Longrightarrow \qquad x'^1 = \frac{1}{2}x^1.}$$

The coordinate compensates for the change in basis so that the underlying vector remains unchanged.

4.5 How the Dual Basis Must Transform

Let

$$f^1, \dots, f^n$$

be the dual basis corresponding to

$$e_1, \dots, e_n.$$

By definition,

$$f^i(e_j) = \delta^i_j.$$

After changing the basis to

$$e'_1, \dots, e'_n,$$

there must be a corresponding new dual basis

$$f'^1, \dots, f'^n$$

satisfying

$$\boxed{f'^i(e'_j) = \delta^i_j.}$$

We already know

$$e'_j = A_{kj} e_k.$$

Suppose the new dual basis has the form

$$f'^i = B_{i\ell} f^\ell.$$

Then

$$\begin{aligned} f'^i(e'_j) &= \left(B_{i\ell} f^\ell\right)\left(A_{kj} e_k\right) \\ &= B_{i\ell} A_{kj} f^\ell(e_k). \end{aligned}$$

Using

$$f^\ell(e_k) = \delta^\ell_k,$$

we obtain

$$f'^i(e'_j) = B_{i\ell} A_{\ell j}.$$

But the defining property of the new dual basis requires

$$f'^i(e'_j) = \delta^i_j.$$

Therefore,

$$B_{i\ell} A_{\ell j} = \delta^i_j.$$

In matrix notation,

$$BA = I.$$

Hence,

$$B = A^{-1}.$$

Therefore, the dual basis transforms according to

$$\boxed{f'^i = (A^{-1})_{ij} f^j.}$$

Thus, just as the vector coordinates compensate for the transformation of the vector basis, the dual basis compensates for the transformation of the original basis.

4.6 The Functional Itself Does Not Change

Now consider a fixed linear functional

$$f \in V^*.$$

Relative to the old dual basis,

$$f = a_i f^i.$$

Relative to the new dual basis,

$$f = a'_i f'^i.$$

Again, these expressions describe the same functional:

$$\boxed{f = a_i f^i = a'_i f'^i.}$$

Substitute the transformation law for the dual basis:

$$f'^i = (A^{-1})_{ij} f^j.$$

Then

$$a_j f^j = a'_i (A^{-1})_{ij} f^j.$$

Since the f^j are linearly independent,

$$a_j = a'_i (A^{-1})_{ij}.$$

Equivalently,

$$\boxed{a'_i = A_{ji}a_j.}$$

In matrix notation, if

$$a = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix},$$

then

$$\boxed{a' = A^T a.}$$

4.7 Why Functional Coordinates Are Called Covariant

Compare the transformation laws:

$$E' = EA,$$

$$x' = A^{-1}x,$$

and

$$a' = A^T a.$$

The vector coordinates transform with the inverse matrix,

$$x' = A^{-1}x,$$

whereas the functional coefficients transform directly with the coefficients of the basis transformation:

$$a'_i = A_{ji}a_j.$$

For this reason, the coefficients

$$a_1, \ldots, a_n$$

of a linear functional are called *covariant coordinates.*

They are conventionally written with lower indices:

$$\boxed{f = a_i f^i.}$$

We therefore arrive at the important distinction

$$\boxed{\begin{array}{lcl} x^i & : & \text{contravariant coordinates of a vector,} \\ a_i & : & \text{covariant coordinates of a functional.} \end{array}}$$

4.8 Why the Scalar Measurement Must Remain Unchanged

We now arrive at the central reason why the two transformation laws must complement each other.

The vector x has not changed.

The functional f has not changed.

Therefore, the number obtained by applying f to x cannot change.

In the original coordinates,

$$f(x) = a_i x^i.$$

In the new coordinates,

$$f(x) = a'_i x'^i.$$

Hence,

$$\boxed{a_i x^i = a'_i x'^i.}$$

Let us verify this directly.

Since

$$a' = A^T a$$

and

$$x' = A^{-1}x,$$

we have

$$\begin{aligned} a'^T x' &= (A^T a)^T (A^{-1}x) \\ &= a^T A(A^{-1}x) \\ &= a^T x. \end{aligned}$$

Therefore,

$$\boxed{f(x) = a^T x = a'^T x'.}$$

In index notation,

$$\boxed{f(x) = a_i x^i = a'_i x'^i.}$$

The covariant and contravariant transformation laws therefore compensate for one another exactly.

4.9 The Measurement Interpretation

This invariance has a simple interpretation.

Suppose a physical quantity has a fixed value. We may change the units or coordinate system used to describe the underlying data, but the physical quantity itself should not change.

Similarly, changing the basis changes the numerical description of both the vector and the functional, but it cannot change the scalar measurement

$$f(x).$$

Thus,

$$\boxed{\text{change of coordinates} \neq \text{change of the underlying object.}}$$

Instead,

$$\boxed{\text{coordinates adjust so that the mathematical relationship remains invariant.}}$$

This principle is considerably broader than linear algebra. It lies behind coordinate transformations throughout geometry, physics, statistics, and tensor analysis.

4.10 A Two-Dimensional Example

Consider the standard basis of $\mathbb{R}^2$,

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Introduce the new basis

$$e_1' = e_1 + e_2, \qquad e_2' = e_2.$$

Then

$$E' = EA,$$

where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Indeed, the first column of A gives

$$e_1' = e_1 + e_2,$$

while the second gives

$$e_2' = e_2.$$

Take the vector

$$x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

in the original basis. Its original coordinate column is

$$[x]_E = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Since

$$A^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix},$$

the new coordinates are

$$\begin{aligned} [x]_{E'} &= A^{-1}[x]_E \\ &= \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 1 \end{bmatrix}. \end{aligned}$$

Indeed,

$$x = 2e_1 + 3e_2$$

and also

$$x = 2e'_1 + e'_2.$$

The coordinates have changed,

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

but the vector has not.

Now define a functional by

$$f(x_1, x_2) = 4x_1 - x_2.$$

Relative to the original basis,

$$a = \begin{bmatrix} 4 \\ -1 \end{bmatrix}.$$

Its new covariant coordinates are

$$\begin{aligned} a' &= A^T a \\ &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ -1 \end{bmatrix}. \end{aligned}$$

Now evaluate the functional using the original coordinates:

$$\begin{aligned} f(x) &= a^T [x]_E \\ &= \begin{bmatrix} 4 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ &= 8 - 3 \\ &= 5. \end{aligned}$$

Using the new coordinates,

$$\begin{aligned} f(x) &= a'^T [x]_{E'} \\ &= \begin{bmatrix} 3 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= 6 - 1 \\ &= 5. \end{aligned}$$

Thus,

$$\boxed{a^T x = a'^T x' = 5.}$$

The numerical coordinates of both objects changed, but their interaction did not.

4.11 Summary: Covariance and Contravariance

Suppose the basis transformation is

$$\boxed{E' = EA.}$$

Then the principal transformation laws developed in this section are

$$\boxed{x' = A^{-1}x}$$

for vector coordinates,

$$\boxed{f'^i = (A^{-1})_{ij} f^j}$$

for the dual basis, and

$$\boxed{a' = A^T a}$$

for functional coordinates.

Thus,

$$\boxed{\begin{array}{lll} x^i & \text{is} & \text{contravariant,} \\ a_i & \text{is} & \text{covariant.} \end{array}}$$

Their opposite transformation behavior guarantees the invariance of the natural pairing:

$$\boxed{f(x) = a_i x^i = a'_i x'^i.}$$

This is the central principle to carry forward:

$$\boxed{\text{coordinate system may change, but the vector, the functional, and the resulting measurement do not.}}$$

So far, the vector space has possessed only its linear structure. We have not yet introduced any intrinsic notion of length, angle, or orthogonality. To obtain these geometric concepts, we must equip the vector space with an inner product. Doing so will reveal a remarkable connection between vectors and linear functionals and will lead directly to the Riesz Representation Theorem.

5 Euclidean Spaces, Inner Products, and the Metric

Up to this point, our vector space V has possessed only its linear structure. We have been able to add vectors, multiply them by scalars, choose bases, and apply linear functionals to them. However, the vector space itself has not yet provided us with any intrinsic notion of length, angle, distance, or orthogonality.

To introduce these geometric concepts, we equip the vector space with an additional structure: an *inner product*.

This additional structure will have another important consequence. It will allow vectors themselves to generate linear functionals, thereby creating a natural bridge between the vector space V and its dual space V^*.

5.1 Inner-Product Spaces and Euclidean Spaces

Definition 5.1 *Let V be a real vector space. An* inner product *on V is a function*

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{R}$$

that assigns a real number

$$\langle x, y \rangle$$

to every pair of vectors $x, y \in V$, and satisfies the following properties.

For all $x, y, z \in V$ and all $\alpha, \beta \in \mathbb{R}$,

1. **Symmetry:**
$$\boxed{\langle x, y \rangle = \langle y, x \rangle.}$$

2. **Linearity:**
$$\boxed{\langle \alpha x + \beta z, y \rangle = \alpha \langle x, y \rangle + \beta \langle z, y \rangle.}$$

3. **Positive definiteness:**
$$\boxed{\langle x, x \rangle \geq 0,}$$
with
$$\boxed{\langle x, x \rangle = 0 \quad \Longleftrightarrow \quad x = 0.}$$

Because we are working over $\mathbb{R}$, symmetry together with linearity in one argument implies linearity in the other argument as well.

A real finite-dimensional vector space equipped with an inner product is commonly called a *Euclidean space.*

Thus, conceptually,

$$\boxed{\text{Euclidean space} = \text{real vector space} + \text{inner product.}}$$

5.2 Geometry Produced by the Inner Product

The inner product introduces geometric information that an abstract vector space does not possess by itself.

The length, or norm, of a vector is defined by

$$\boxed{\|x\| = \sqrt{\langle x, x\rangle}.}$$

Two vectors are orthogonal when

$$\boxed{\langle x, y\rangle = 0.}$$

For nonzero vectors, the angle θ between them is determined by

$$\boxed{\cos\theta = \frac{\langle x, y\rangle}{\|x\|\|y\|}.}$$

Thus, an inner product allows us to discuss

$$\text{length}, \qquad \text{angle}, \qquad \text{orthogonality}, \qquad \text{distance.}$$

The distance between two vectors is simply

$$\boxed{d(x, y) = \|x - y\|.}$$

These geometric concepts are not consequences of the vector-space axioms alone. They arise from the additional inner-product structure.

5.3 The Standard Euclidean Inner Product

The most familiar example occurs in

$$V = \mathbb{R}^n.$$

For

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix},$$

the standard Euclidean inner product is

$$\boxed{\langle x, y \rangle = x^T y = \sum_{i=1}^{n} x_i y_i.}$$

For example, if

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

then

$$\langle x, y \rangle = 1(3) + 2(4) = 11.$$

The norm of x is therefore

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

5.4 A Vector Generates a Linear Functional

We now arrive at the crucial connection between Euclidean spaces and dual spaces.

Fix a vector

$$y \in V.$$

Define a function

$$f_y : V \longrightarrow \mathbb{R}$$

by

$$\boxed{f_y(x) = \langle x, y \rangle.}$$

Since y is fixed, the only variable is x.

We now verify that f_y is a linear functional. For $x, z \in V$ and $\alpha, \beta \in \mathbb{R}$,

$$\begin{aligned} f_y(\alpha x + \beta z) &= \langle \alpha x + \beta z, y \rangle \\ &= \alpha \langle x, y \rangle + \beta \langle z, y \rangle \\ &= \alpha f_y(x) + \beta f_y(z). \end{aligned}$$

Therefore,

$$\boxed{f_y \in V^*.}$$

Thus, every vector y in a Euclidean space naturally generates a linear functional:

$$\boxed{y \in V \quad \longmapsto \quad f_y \in V^*, \qquad f_y(x) = \langle x, y \rangle.}$$

This is the first indication that the inner product establishes a relationship between V and V^*.

5.5 A Simple Example

Consider

$$V = \mathbb{R}^2$$

with the standard Euclidean inner product, and let

$$y = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

Then y defines the functional

$$f_y(x) = \langle x, y \rangle.$$

For

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we obtain

$$\begin{aligned} f_y(x) &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \\ &= 3x_1 + 5x_2. \end{aligned}$$

Thus, the familiar linear functional

$$f(x) = 3x_1 + 5x_2$$

can be represented by taking the inner product with the vector

$$y = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

This immediately raises the reverse question:

> If every vector generates a linear functional through the inner product, does every linear functional arise from some vector in this way?

We will answer this question formally with the Riesz Representation Theorem. Before doing so, however, we must understand how the inner product behaves in an arbitrary, possibly non-orthonormal, basis.

5.6 Orthonormal Bases

A basis

$$e_1, \ldots, e_n$$

is called *orthonormal* if

$$\boxed{\langle e_i, e_j \rangle = \delta_{ij}.}$$

Thus,

$$\langle e_i, e_i \rangle = 1$$

for every i, while

$$\langle e_i, e_j \rangle = 0 \qquad \text{when } i \neq j.$$

Therefore, each basis vector has unit length and distinct basis vectors are mutually orthogonal.

In an orthonormal basis, if

$$x = x^i e_i, \qquad y = y^j e_j,$$

then

$$\begin{aligned} \langle x, y \rangle &= \langle x^i e_i, y^j e_j \rangle \\ &= x^i y^j \langle e_i, e_j \rangle \\ &= x^i y^j \delta_{ij}. \end{aligned}$$

The Kronecker delta selects the terms for which $i = j$, giving

$$\boxed{\langle x, y \rangle = \sum_{i=1}^{n} x^i y^i.}$$

This is precisely the familiar dot-product formula.

The simplicity of this expression is therefore a consequence of the basis being orthonormal.

5.7 The Metric Coefficients

Suppose now that

$$e_1, \ldots, e_n$$

is an arbitrary basis. The basis vectors need not have unit length and need not be mutually orthogonal.

We define the *metric coefficients* by

$$\boxed{g_{ij} = \langle e_i, e_j \rangle.}$$

Collecting these numbers produces the matrix

$$\boxed{G = [g_{ij}].}$$

Explicitly,

$$G = \begin{bmatrix} \langle e_1, e_1 \rangle & \langle e_1, e_2 \rangle & \cdots & \langle e_1, e_n \rangle \\ \langle e_2, e_1 \rangle & \langle e_2, e_2 \rangle & \cdots & \langle e_2, e_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle e_n, e_1 \rangle & \langle e_n, e_2 \rangle & \cdots & \langle e_n, e_n \rangle \end{bmatrix}.$$

This matrix is also called the *Gram matrix* of the basis.

Because the inner product is symmetric,

$$g_{ij} = g_{ji},$$

and therefore

$$\boxed{G^T = G.}$$

5.8 The Geometric Meaning of g_{ij}

The metric coefficients encode the geometry of the chosen basis.

The diagonal entries satisfy

$$g_{ii} = \langle e_i, e_i \rangle = \|e_i\|^2.$$

Thus,

$$\boxed{g_{ii} = \|e_i\|^2.}$$

For $i \neq j$,

$$g_{ij} = \langle e_i, e_j \rangle = \|e_i\| \|e_j\| \cos \theta_{ij},$$

where θ_{ij} is the angle between e_i and e_j.

Hence the matrix G records both the lengths of the basis vectors and their mutual angular relationships.

If the basis is orthonormal, then

$$g_{ij} = \delta_{ij},$$

so

$$\boxed{G = I.}$$

The familiar Euclidean dot-product formulas are therefore the special case in which the metric matrix is the identity.

5.9 Inner Products in an Arbitrary Basis

Let

$$x = x^i e_i, \qquad y = y^j e_j.$$

Then

$$\begin{aligned} \langle x, y \rangle &= \langle x^i e_i, y^j e_j \rangle \\ &= x^i y^j \langle e_i, e_j \rangle. \end{aligned}$$

Since

$$\langle e_i, e_j \rangle = g_{ij},$$

we obtain

$$\boxed{\langle x, y \rangle = g_{ij} x^i y^j.}$$

In matrix notation,

$$\boxed{\langle x, y \rangle = x^T G y,}$$

where x and y now denote the coordinate columns relative to the chosen basis.

If the basis is orthonormal,

$$G = I,$$

and therefore

$$x^T G y = x^T y.$$

Thus, the matrix G corrects for the geometry of a non-orthonormal coordinate system.

5.10 From a Vector to Covariant Coordinates

Let

$$y = y^j e_j.$$

As shown earlier, y generates the functional

$$f_y(x) = \langle x, y \rangle.$$

The covariant coordinates of this functional are its values on the basis vectors:

$$a_i = f_y(e_i).$$

Therefore,

$$a_i = \langle e_i, y \rangle.$$

Substituting

$$y = y^j e_j,$$

we obtain

$$\begin{aligned} a_i &= \langle e_i, y^j e_j \rangle \\ &= y^j \langle e_i, e_j \rangle \\ &= g_{ij} y^j. \end{aligned}$$

Hence,

$$\boxed{a_i = g_{ij} y^j.}$$

Because the functional is generated by the vector y, it is common to denote these covariant coordinates by y_i rather than a_i. Thus,

$$\boxed{y_i = g_{ij} y^j.}$$

This operation converts the contravariant coordinates y^j of a vector into the covariant coordinates y_i of its associated functional.

5.11 Lowering an Index

The operation

$$\boxed{y_i = g_{ij} y^j}$$

is called *lowering an index*.

In matrix notation,

$$\boxed{y_{\text{cov}} = G\, y_{\text{contra}}.}$$

Thus, the metric matrix acts as the coordinate mechanism that converts a vector into the coordinate representation of the corresponding linear functional.

Schematically,

$$\boxed{y^j \xrightarrow{\;g_{ij}\;} y_i.}$$

If the basis is orthonormal,

$$g_{ij} = \delta_{ij},$$

and therefore

$$y_i = \delta_{ij} y^j.$$

Using the selection property of the Kronecker delta,

$$y_i = y^i$$

numerically.

This explains why the distinction between covariant and contravariant coordinates is almost invisible in ordinary orthonormal Cartesian coordinates.

5.12 The Inverse Metric

Because

$$e_1, \dots, e_n$$

is a basis and the inner product is positive definite, the Gram matrix G is positive definite and therefore invertible.

We denote the entries of its inverse by

$$\boxed{G^{-1} = [g^{ij}].}$$

Thus,

$$[g^{ij}][g_{jk}] = I.$$

In index notation,

$$\boxed{g^{ij} g_{jk} = \delta^i_k.}$$

This is precisely the indexed version of

$$G^{-1}G = I.$$

The Kronecker delta introduced earlier therefore reappears naturally as the identity connecting the metric and its inverse.

5.13 Raising an Index

Starting from

$$y_i = g_{ij}y^j,$$

multiply by g^{ki}:

$$g^{ki}y_i = g^{ki}g_{ij}y^j.$$

Since

$$g^{ki}g_{ij} = \delta^k_j,$$

we obtain

$$g^{ki}y_i = \delta^k_j y^j.$$

Using the selection property,

$$\delta^k_j y^j = y^k.$$

Therefore,

$$\boxed{y^k = g^{ki}y_i.}$$

Renaming the dummy indices gives the familiar form

$$\boxed{y^i = g^{ij}y_j.}$$

This operation is called *raising an index*.

Thus,

$$\boxed{\begin{array}{ccc} y^j & \xrightarrow{g_{ij}} & y_i \\ y^i & \xleftarrow{g^{ij}} & y_j. \end{array}}$$

In matrix notation,

$$y_{\text{cov}} = G y_{\text{contra}},$$

and

$$y_{\text{contra}} = G^{-1} y_{\text{cov}}.$$

5.14 The Inner Product as a Natural Pairing

Recall that

$$\langle x, y \rangle = g_{ij} x^i y^j.$$

But

$$y_i = g_{ij} y^j.$$

Therefore,

$$\boxed{\langle x, y \rangle = x^i y_i.}$$

Compare this with the natural pairing between a functional and a vector:

$$f(x) = a_i x^i.$$

The expressions have exactly the same form:

$$\boxed{f(x) = a_i x^i,}$$

and

$$\boxed{\langle x, y \rangle = y_i x^i.}$$

This is not an accidental similarity.

The inner product has converted the vector y, whose contravariant coordinates are y^i, into a functional whose covariant coordinates are

$$y_i = g_{ij} y^j.$$

Thus,

$$\boxed{\text{inner product} \quad \Longrightarrow \quad \text{vector } y \text{ determines a functional } f_y.}$$

5.15 The Bridge Between V and V^*

We may now summarize the construction.

Start with

$$y \in V.$$

The inner product defines

$$f_y(x) = \langle x, y \rangle.$$

Thus,

$$y \longmapsto f_y.$$

In coordinates,

$$y^j \longmapsto y_i = g_{ij} y^j.$$

Hence the metric matrix provides the coordinate realization of the map

$$\boxed{V \longrightarrow V^*.}$$

The inverse metric performs the reverse coordinate conversion:

$$\boxed{y^i = g^{ij} y_j.}$$

This leads to the central question:

$$\boxed{\text{Does every linear functional in } V^* \text{ arise from exactly one vector in } V \text{ through the inner product?}}$$

In finite-dimensional Euclidean spaces, the answer is yes.

That result is the *Riesz Representation Theorem*, to which we now turn.

5.16 Summary

Equipping a real vector space with an inner product introduces geometric structure and simultaneously creates a connection between vectors and linear functionals.

For an arbitrary basis,

$$\boxed{g_{ij} = \langle e_i, e_j \rangle}$$

defines the metric coefficients, and

$$\boxed{G = [g_{ij}]}$$

is the Gram, or metric, matrix.

The inner product takes the coordinate form

$$\boxed{\langle x, y \rangle = g_{ij} x^i y^j = x^i y_i.}$$

The metric lowers indices,

$$\boxed{y_i = g_{ij} y^j,}$$

while the inverse metric raises them,

$$\boxed{y^i = g^{ij} y_j,}$$

with

$$\boxed{g^{ij} g_{jk} = \delta^i_k.}$$

Most importantly, every vector $y \in V$ generates a linear functional

$$\boxed{f_y(x) = \langle x, y \rangle.}$$

We are therefore ready to determine whether this correspondence accounts for every element of the dual space V^*.

6 The Riesz Representation Theorem

The previous section established that every vector $y \in V$ in a finite-dimensional Euclidean space generates a linear functional

$$f_y(x) = \langle x, y \rangle.$$

Thus, the inner product defines a map

$$V \longrightarrow V^*.$$

The natural question is whether this map reaches every element of the dual space. Equivalently, given an arbitrary linear functional

$$f \in V^*,$$

can we always find a vector $y \in V$ such that

$$f(x) = \langle x, y \rangle$$

for every $x \in V$?

In finite-dimensional Euclidean spaces, the answer is yes, and the representing vector is unique.

6.1 Statement of the Theorem

Theorem 6.1 (Finite-Dimensional Riesz Representation Theorem) *Let V be a finite-dimensional real inner-product space. Then for every linear functional*

$$f \in V^*,$$

there exists a unique vector

$$y \in V$$

such that

$$\boxed{f(x) = \langle x, y \rangle \qquad \textit{for every } x \in V.}$$

The theorem contains two separate claims:

$$\boxed{\text{existence}}$$

and

$$\boxed{\text{uniqueness.}}$$

Existence means that at least one representing vector y can be found. Uniqueness means that no two different vectors can represent the same functional through the inner product.

6.2 Proof of Existence Using an Orthonormal Basis

Let

$$e_1, \ldots, e_n$$

be an orthonormal basis of V. Thus,

$$\langle e_i, e_j \rangle = \delta_{ij}.$$

Let

$$f \in V^*$$

be an arbitrary linear functional.

Because f is completely determined by its values on the basis, define

$$\boxed{a_i = f(e_i).}$$

We now construct the vector

$$\boxed{y = a_i e_i.}$$

Written explicitly,

$$y = a_1 e_1 + \cdots + a_n e_n.$$

We will show that this vector satisfies

$$f(x) = \langle x, y \rangle$$

for every $x \in V$.

Take an arbitrary vector

$$x = x^j e_j.$$

By linearity of f,

$$\begin{aligned} f(x) &= f(x^j e_j) \\ &= x^j f(e_j). \end{aligned}$$

Since

$$f(e_j) = a_j,$$

we obtain

$$\boxed{f(x) = x^j a_j.}$$

Now calculate the inner product with the candidate vector y:

$$\begin{aligned}\langle x, y \rangle &= \left\langle x^j e_j, a_i e_i \right\rangle \\ &= x^j a_i \langle e_j, e_i \rangle.\end{aligned}$$

Because the basis is orthonormal,

$$\langle e_j, e_i \rangle = \delta_{ji}.$$

Therefore,

$$\begin{aligned}\langle x, y \rangle &= x^j a_i \delta_{ji} \\ &= x^j a_j.\end{aligned}$$

Hence,

$$\boxed{\langle x, y \rangle = f(x)}$$

for every $x \in V$.

Therefore, a representing vector exists.

6.3 Proof of Uniqueness

Suppose that two vectors

$$y, z \in V$$

both represent the same functional f. Then

$$f(x) = \langle x, y \rangle$$

and

$$f(x) = \langle x, z \rangle$$

for every $x \in V$.

Therefore,

$$\langle x, y \rangle = \langle x, z \rangle$$

for every x.

Subtracting gives

$$\langle x, y - z \rangle = 0$$

for every $x \in V$.

Now choose

$$x = y - z.$$

Then

$$\langle y - z, y - z \rangle = 0.$$

By positive definiteness of the inner product,

$$\langle v, v \rangle = 0 \quad \Longrightarrow \quad v = 0.$$

Therefore,

$$y - z = 0,$$

and hence

$$\boxed{y = z.}$$

Thus, the representing vector is unique.

6.4 Interpretation of the Theorem

The theorem establishes a one-to-one correspondence

$$\boxed{V \longleftrightarrow V^*.}$$

Specifically,

$$y \in V$$

corresponds to the functional

$$f_y(x) = \langle x, y \rangle.$$

Conversely, every functional

$$f \in V^*$$

is represented by exactly one vector $y \in V$.

Thus,

$$\boxed{f(x) = \langle x, y \rangle}$$

is not merely one way to construct linear functionals; in a finite-dimensional Euclidean space, it describes every possible linear functional.

6.5 The Riesz Map

The correspondence may be written as a map

$$\mathcal{R} : V \longrightarrow V^*$$

defined by

$$\boxed{\mathcal{R}(y) = f_y, \qquad f_y(x) = \langle x, y \rangle.}$$

This map is often called the *Riesz map*.

In the finite-dimensional real case, the Riesz Representation Theorem says that

$$\mathcal{R}$$

is a bijection.

Thus,

$$\boxed{\mathcal{R} : V \xrightarrow{\sim} V^*.}$$

The symbol

$$\sim$$

indicates that the two spaces are naturally identified through the inner product.

It is important, however, to remember the conceptual distinction:

$$\boxed{V \neq V^* \text{ as collections of objects.}}$$

Vectors and linear functionals remain different kinds of objects. The inner product provides a natural one-to-one correspondence between them.

6.6 Riesz Representation in a Non-Orthonormal Basis

The orthonormal-basis proof is especially simple because

$$\langle e_i, e_j \rangle = \delta_{ij}.$$

In a general basis, however,

$$\langle e_i, e_j \rangle = g_{ij}.$$

Suppose

$$f \in V^*$$

has covariant coordinates

$$\boxed{a_i = f(e_i).}$$

Let the representing vector be

$$y = y^j e_j.$$

Since

$$f(x) = \langle x, y \rangle,$$

we may evaluate both sides at $x = e_i$:

$$f(e_i) = \langle e_i, y \rangle.$$

Therefore,

$$a_i = \langle e_i, y \rangle.$$

Substituting

$$y = y^j e_j,$$

we obtain

$$\begin{aligned} a_i &= \langle e_i, y^j e_j \rangle \\ &= y^j \langle e_i, e_j \rangle \\ &= g_{ij} y^j. \end{aligned}$$

Hence,

$$\boxed{a_i = g_{ij} y^j.}$$

In matrix notation,

$$\boxed{a = Gy.}$$

Thus, the metric matrix converts the contravariant coordinates of the Riesz vector into the covariant coordinates of the functional.

6.7 Recovering the Representing Vector

Since

$$a = Gy,$$

and G is invertible,

$$y = G^{-1}a.$$

Using

$$G^{-1} = [g^{ij}],$$

we obtain

$$\boxed{y^i = g^{ij}a_j.}$$

This formula gives the coordinates of the unique Riesz vector directly from the coordinates of the functional.

Thus, the complete coordinate correspondence is

$$\boxed{a_i = g_{ij}y^j,}$$

and

$$\boxed{y^i = g^{ij}a_j.}$$

These are precisely the index-lowering and index-raising operations developed earlier.

6.8 The Orthonormal Basis as a Special Case

If the basis is orthonormal, then

$$g_{ij} = \delta_{ij},$$

and therefore

$$G = I.$$

Hence,

$$a_i = \delta_{ij} y^j,$$

so

$$a_i = y^i$$

numerically.

Likewise,

$$G^{-1} = I,$$

and

$$y^i = a_i.$$

Thus, in an orthonormal basis, the coordinates of the functional and the coordinates of its Riesz vector appear numerically identical.

This explains why the distinction between vectors and covectors is often hidden in ordinary Cartesian coordinates.

6.9 A Simple Example

Consider

$$V = \mathbb{R}^3$$

with the standard Euclidean inner product, and define

$$f(x_1, x_2, x_3) = 2x_1 - 3x_2 + 5x_3.$$

The standard basis is orthonormal, so the functional coordinates are

$$a = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}.$$

Since

$$G = I,$$

the Riesz vector has the same numerical coordinates:

$$y = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}.$$

Indeed,

$$\begin{aligned} \langle x, y \rangle &= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} \\ &= 2x_1 - 3x_2 + 5x_3 \\ &= f(x). \end{aligned}$$

Thus,

$$\boxed{f(x) = \langle x, y \rangle.}$$

6.10 A Data-Science Interpretation

The Riesz Representation Theorem provides a useful way to interpret linear measurements in Euclidean data spaces.

Suppose a data point is represented by

$$x \in \mathbb{R}^n,$$

and a linear score is given by

$$f(x).$$

The theorem tells us that there is a unique vector y such that

$$\boxed{f(x) = \langle x, y \rangle.}$$

Thus, every linear scalar measurement may be interpreted geometrically as an inner product with a particular direction.

In standard Euclidean coordinates,

$$f(x) = y^T x.$$

Therefore, familiar expressions such as

$$w^T x$$

in linear regression, linear classification, and projection-based methods are concrete instances of the Riesz correspondence.

The vector w is not merely a list of coefficients. It is the vector representing a linear functional through the inner product.

In a non-orthonormal coordinate system, the relationship becomes

$$a = Gy,$$

so the numerical coefficient vector associated with the functional and the coordinate vector of the representing direction need not be identical.

This distinction becomes important whenever the geometry of the feature space is not represented by the identity metric.

6.11 Finite and Infinite Dimensions

The theorem developed here concerns finite-dimensional Euclidean spaces.

In finite dimensions, every linear functional is automatically continuous, so the theorem applies to every element of V^*.

In an infinite-dimensional Hilbert space, the corresponding theorem requires the functional to be continuous, or equivalently bounded. The infinite-dimensional version states that every continuous linear functional can be represented uniquely through an inner product.

That broader theory belongs to functional analysis. The finite-dimensional version developed here, however, already contains the essential geometric idea:

$$\boxed{\text{linear measurement} \quad \longleftrightarrow \quad \text{inner product with a unique vector.}}$$

6.12 Summary

The Riesz Representation Theorem completes the bridge between vectors and linear functionals in finite-dimensional Euclidean spaces.

For every

$$f \in V^*,$$

there exists a unique

$$y \in V$$

such that

$$\boxed{f(x) = \langle x, y \rangle}$$

for every $x \in V$.

In an arbitrary basis, if

$$a_i = f(e_i),$$

then the Riesz vector satisfies

$$\boxed{a_i = g_{ij} y^j,}$$

and therefore

$$\boxed{y^i = g^{ij} a_j.}$$

Thus, the metric and inverse metric provide the coordinate realization of the Riesz correspondence:

$$\boxed{V \longleftrightarrow V^*.}$$

The next section will bring all of these ideas together in a complete three-dimensional example using a non-orthonormal basis.

7 A Complete Three-Dimensional Worked Example

The previous sections introduced several ideas in rapid succession: dual spaces, covariant and contravariant coordinates, non-orthonormal bases, metric coefficients, raising and lowering indices, and the Riesz representation of a linear functional.

It is entirely reasonable if these ideas do not yet feel completely natural.

Concepts involving duality, coordinate transformations, and upper and lower indices usually become clear only after they have been used repeatedly. Very few readers become comfortable with tensor-style index notation, covariance, or duality simply by hearing a lecture or reading a set of notes once. These ideas are learned by calculation, by seeing the same structure appear in several different forms, and by returning to the theory after working through concrete examples.

Therefore, even if some parts of the previous sections remain unclear, the reader is encouraged to continue through the present example carefully.

The purpose of this section is not merely to obtain an answer. Rather, it is to watch the complete chain

$$\boxed{\text{functional} \longrightarrow \text{covariant coordinates} \longrightarrow \text{metric} \longrightarrow \text{contravariant coordinates} \longrightarrow \text{Riesz vector}}$$

operate in an explicit three-dimensional Euclidean space.

After completing the example, a second reading of the preceding sections should make many of the abstract formulas considerably more transparent.

7.1 The Problem

Let

$$V = \mathbb{R}^3$$

with the standard Euclidean inner product

$$\boxed{\langle x, y \rangle = x^T y.}$$

We will not, however, use the standard Cartesian basis as our working basis.

Instead, consider the non-orthonormal basis

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad e_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Now, let a linear functional $f \in V^*$ is specified by its values on these basis vectors:

$$\boxed{f(e_1) = 2, \qquad f(e_2) = 5, \qquad f(e_3) = 1.}$$

Our objective is to find the unique vector

$$y \in V$$

such that

$$\boxed{f(x) = \langle x, y \rangle \qquad \text{for every } x \in V.}$$

By the Riesz Representation Theorem, such a vector exists and is unique.

We will determine it step by step.

7.2 Step 1: Identify the Covariant Coordinates of the Functional

The covariant coordinates of a functional are obtained by evaluating the functional on the basis vectors:

$$\boxed{a_i = f(e_i).}$$

Therefore,

$$a_1 = f(e_1) = 2,$$

$$a_2 = f(e_2) = 5,$$

and

$$a_3 = f(e_3) = 1.$$

Hence,

$$\boxed{a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}.}$$

These numbers describe the functional relative to the chosen non-orthonormal basis.

7.3 Step 2: Construct the Metric Matrix

The metric coefficients are defined by

$$\boxed{g_{ij} = \langle e_i, e_j \rangle.}$$

Thus, we must compute every pairwise inner product between the basis vectors.

We begin with

$$g_{11} = \langle e_1, e_1 \rangle.$$

Since

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

we obtain

$$\begin{aligned} g_{11} &= 1(1) + 0(0) + 0(0) \\ &= 1. \end{aligned}$$

Next,

$$g_{12} = \langle e_1, e_2 \rangle.$$

Therefore,

$$\begin{aligned} g_{12} &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ &= 1(1) + 0(1) + 0(0) \\ &= 1. \end{aligned}$$

Similarly,

$$\begin{aligned} g_{13} &= \langle e_1, e_3 \rangle \\ &= 1(0) + 0(1) + 0(1) \\ &= 0. \end{aligned}$$

For the second basis vector,

$$\begin{aligned} g_{21} &= \langle e_2, e_1 \rangle \\ &= 1, \end{aligned}$$

$$\begin{aligned} g_{22} &= \langle e_2, e_2 \rangle \\ &= 1^2 + 1^2 + 0^2 \\ &= 2, \end{aligned}$$

and

$$\begin{aligned} g_{23} &= \langle e_2, e_3 \rangle \\ &= 1(0) + 1(1) + 0(1) \\ &= 1. \end{aligned}$$

Finally,

$$\begin{aligned} g_{31} &= \langle e_3, e_1 \rangle \\ &= 0, \end{aligned}$$

$$\begin{aligned} g_{32} &= \langle e_3, e_2 \rangle \\ &= 1, \end{aligned}$$

and

$$\begin{aligned} g_{33} &= \langle e_3, e_3 \rangle \\ &= 0^2 + 1^2 + 1^2 \\ &= 2. \end{aligned}$$

Therefore,

$$\boxed{G = [g_{ij}] = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.}$$

Notice that

$$G^T = G,$$

as expected from symmetry of the inner product.

7.4 Step 3: Relate the Functional to the Riesz Vector

Let the Riesz vector have contravariant coordinates

$$y^1, \qquad y^2, \qquad y^3.$$

Thus,

$$\boxed{y = y^1 e_1 + y^2 e_2 + y^3 e_3.}$$

From the Riesz correspondence,

$$f(x) = \langle x, y \rangle.$$

In particular,

$$a_i = f(e_i) = \langle e_i, y \rangle.$$

Since

$$y = y^j e_j,$$

we have

$$\begin{aligned} a_i &= \langle e_i, y^j e_j \rangle \\ &= y^j \langle e_i, e_j \rangle \\ &= g_{ij} y^j. \end{aligned}$$

Hence,

$$\boxed{a_i = g_{ij} y^j.}$$

In matrix form,

$$\boxed{a = Gy.}$$

For our problem,

$$\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} y^1 \\ y^2 \\ y^3 \end{bmatrix}.$$

7.5 Step 4: Solve the Linear System

Expanding the matrix equation gives

$$y^1 + y^2 = 2,$$

$$y^1 + 2y^2 + y^3 = 5,$$

and

$$y^2 + 2y^3 = 1.$$

From the first equation,

$$\boxed{y^1 = 2 - y^2.}$$

From the third equation,

$$2y^3 = 1 - y^2,$$

so

$$\boxed{y^3 = \frac{1 - y^2}{2}.}$$

Substitute these expressions into

$$y^1 + 2y^2 + y^3 = 5.$$

We obtain

$$(2 - y^2) + 2y^2 + \frac{1 - y^2}{2} = 5.$$

Combine the first two terms:

$$2 + y^2 + \frac{1 - y^2}{2} = 5.$$

Multiply by 2:

$$4 + 2y^2 + 1 - y^2 = 10.$$

Therefore,

$$5 + y^2 = 10.$$

Hence,

$$\boxed{y^2 = 5.}$$

Then

$$y^1 = 2 - 5 = -3,$$

so

$$\boxed{y^1 = -3.}$$

Finally,

$$y^3 = \frac{1-5}{2} = -2.$$

Therefore,

$$\boxed{y^3 = -2.}$$

Thus, the contravariant coordinates of the Riesz vector relative to our non-orthonormal basis are

$$\boxed{[y]_E = \begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix}.}$$

Equivalently,

$$\boxed{y = -3e_1 + 5e_2 - 2e_3.}$$

7.6 Step 5: Convert the Riesz Vector to Cartesian Coordinates

The coordinates

$$\begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix}$$

are coordinates relative to the basis

$$e_1, e_2, e_3.$$

They are not yet the standard Cartesian coordinates of the vector.

Substitute the basis vectors:

$$y = -3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Compute each contribution:

$$-3e_1 = \begin{bmatrix} -3 \\ 0 \\ 0 \end{bmatrix},$$

$$5e_2 = \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix},$$

and

$$-2e_3 = \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} y &= \begin{bmatrix} -3 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}. \end{aligned}$$

Hence the actual Riesz vector in standard Cartesian coordinates is

$$\boxed{y = \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}.}$$

7.7 Step 6: Recover the Functional in Cartesian Coordinates

The Riesz theorem states that

$$f(x) = \langle x, y \rangle.$$

Let an arbitrary Cartesian vector be

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Since

$$y = \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix},$$

we obtain

$$\begin{aligned} f(x) &= \langle x, y \rangle \\ &= x^T y \\ &= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \\ &= 2x_1 + 3x_2 - 2x_3. \end{aligned}$$

Therefore,

$$\boxed{f(x_1, x_2, x_3) = 2x_1 + 3x_2 - 2x_3.}$$

7.8 Step 7: Verify the Original Measurements

We were originally told that

$$f(e_1) = 2, \qquad f(e_2) = 5, \qquad f(e_3) = 1.$$

We should verify that the recovered functional satisfies all three conditions.

For

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

we have

$$\begin{aligned} f(e_1) &= 2(1) + 3(0) - 2(0) \\ &= 2. \end{aligned}$$

Thus,

$$\boxed{f(e_1) = 2.}$$

Next,

$$e_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} f(e_2) &= 2(1) + 3(1) - 2(0) \\ &= 5. \end{aligned}$$

Hence,

$$\boxed{f(e_2) = 5.}$$

Finally,

$$e_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} f(e_3) &= 2(0) + 3(1) - 2(1) \\ &= 1. \end{aligned}$$

Thus,

$$\boxed{f(e_3) = 1.}$$

All three original measurements are recovered exactly.

7.9 Step 8: Compute the Inverse Metric

Instead of solving

$$Gy = a$$

directly, we may use the inverse metric.

Recall

$$G = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Its inverse is

$$\boxed{G^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}.}$$

We denote the entries of this matrix by

$$G^{-1} = [g^{ij}].$$

The Riesz vector coordinates satisfy

$$\boxed{y^i = g^{ij} a_j.}$$

In matrix form,

$$\boxed{y = G^{-1} a.}$$

Therefore,

$$y = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}.$$

Compute the first component:

$$\begin{aligned} y^1 &= 3(2) - 2(5) + 1(1) \\ &= 6 - 10 + 1 \\ &= -3. \end{aligned}$$

The second component is

$$\begin{aligned} y^2 &= -2(2) + 2(5) - 1(1) \\ &= -4 + 10 - 1 \\ &= 5. \end{aligned}$$

The third component is

$$\begin{aligned} y^3 &= 1(2) - 1(5) + 1(1) \\ &= 2 - 5 + 1 \\ &= -2. \end{aligned}$$

Hence,

$$\boxed{y = \begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix}_E.}$$

This agrees exactly with the result obtained by solving the linear system.

7.10 Step 9: Verify Raising and Lowering of Indices

The covariant coordinates of the functional are

$$a_i = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix},$$

while the contravariant coordinates of the representing vector are

$$y^i = \begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix}.$$

The lowering relation is

$$\boxed{a_i = g_{ij} y^j.}$$

In matrix notation,

$$a = Gy.$$

Verify:

$$\begin{aligned} Gy &= \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} -3+5 \\ -3+10-2 \\ 5-4 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\boxed{a_i = g_{ij} y^j.}$$

Conversely, raising the index gives

$$\boxed{y^i = g^{ij} a_j.}$$

Thus the metric and inverse metric provide two inverse coordinate operations:

$$\boxed{y^j \xrightarrow{g_{ij}} a_i}$$

and

$$\boxed{a_j \xrightarrow{g^{ij}} y^i.}$$

7.11 Step 10: Construct the Reciprocal Basis

Although the Riesz vector has already been determined, the inverse metric also allows us to construct another useful set of vectors.

Define

$$\boxed{e^i = g^{ij} e_j.}$$

These vectors satisfy

$$\boxed{\langle e^i, e_j \rangle = \delta^i_j.}$$

They are called the *reciprocal basis vectors* associated with the basis e_1, e_2, e_3.

Using the rows of G^{-1},

$$G^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix},$$

we obtain

$$e^1 = 3e_1 - 2e_2 + e_3.$$

Substituting,

$$\begin{aligned} e^1 &= 3\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 2\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\boxed{e^1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.}$$

Similarly,

$$e^2 = -2e_1 + 2e_2 - e_3,$$

so

$$\begin{aligned} e^2 &= -2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}. \end{aligned}$$

Hence,

$$\boxed{e^2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.}$$

Finally,

$$e^3 = e_1 - e_2 + e_3,$$

which gives

$$\begin{aligned} e^3 &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\boxed{e^3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.}$$

7.12 Step 11: Verify the Reciprocal Basis

We should verify at least one complete row of the defining relation

$$\langle e^i, e_j \rangle = \delta^i_j.$$

For e^1,

$$e^1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

First,

$$\begin{aligned} \langle e^1, e_1 \rangle &= \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ &= 1. \end{aligned}$$

Next,

$$\begin{aligned} \langle e^1, e_2 \rangle &= \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ &= 1 - 1 \\ &= 0. \end{aligned}$$

Finally,

$$\begin{aligned} \langle e^1, e_3 \rangle &= \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \\ &= -1 + 1 \\ &= 0. \end{aligned}$$

Therefore,

$$\boxed{\langle e^1, e_j \rangle = \delta^1_j.}$$

The same property holds for e^2 and e^3.

7.13 Step 12: Recover the Riesz Vector from the Reciprocal Basis

The covariant coordinates of the functional are

$$a_1 = 2, \qquad a_2 = 5, \qquad a_3 = 1.$$

The Riesz vector may be written directly as

$$\boxed{y = a_i e^i.}$$

Thus,

$$y = 2e^1 + 5e^2 + e^3.$$

Substituting the reciprocal basis vectors,

$$\begin{aligned} y &= 2 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 5 \\ -5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}. \end{aligned}$$

Again,

$$\boxed{y = \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}.}$$

Thus, every route leads to the same unique Riesz vector.

7.14 What the Example Has Shown

It is useful to review the complete calculation without interruption.

We began with the functional measurements

$$\boxed{a_i = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}.}$$

The geometry of the non-orthonormal basis was encoded by

$$\boxed{G = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.}$$

The Riesz vector coordinates were obtained from

$$a = Gy,$$

or equivalently,

$$y = G^{-1}a.$$

This gave

$$\boxed{[y]_E = \begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix}.}$$

Therefore,

$$y = -3e_1 + 5e_2 - 2e_3.$$

Converting to standard Cartesian coordinates produced

$$\boxed{y = \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}.}$$

The corresponding functional is

$$\boxed{f(x) = \langle x, y \rangle = 2x_1 + 3x_2 - 2x_3.}$$

Finally,

$$f(e_1) = 2, \qquad f(e_2) = 5, \qquad f(e_3) = 1,$$

exactly as required.

7.15 The Entire Structure in One Diagram

The complete process may be summarized as

$$\boxed{\begin{array}{ccccc} \text{functional} & & \text{covariant coordinates} & & \text{Riesz vector coordinates} \\ f & \longrightarrow & a_i = f(e_i) & \xrightarrow{g^{ij}} & y^i \\ & & \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} & & \begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix} \end{array}}$$

with

$$y^i = g^{ij} a_j.$$

The reverse direction is

$$\boxed{a_i = g_{ij} y^j.}$$

The coordinates y^i then reconstruct the vector

$$y = y^i e_i,$$

and the functional is recovered through

$$\boxed{f(x) = \langle x, y \rangle.}$$

7.16 A Final Remark for the Reader

If the distinction among

$$x^i, \qquad x_i, \qquad g_{ij}, \qquad g^{ij},$$

still requires conscious thought, that is not a sign that the discussion has failed. These ideas are inherently relational: their meaning becomes clear through repeated use.

A first reading usually establishes familiarity.

A worked example establishes mechanics.

A second reading begins to reveal structure.

Only after several encounters do covariance, contravariance, duality, and index manipulation begin to feel natural.

The important point at this stage is to recognize the underlying pattern:

$$\boxed{\text{the mathematical objects remain fixed, while their coordinate descriptions depend on the chosen basis.}}$$

The metric provides the bridge between vector coordinates and functional coordinates, and the Riesz Representation Theorem tells us that, in a finite-dimensional Euclidean space, every linear functional can be viewed as an inner-product measurement against one unique vector.

The reader is therefore encouraged to return to the earlier sections after working through this example. Much of the notation that may initially have appeared abstract should now have a concrete computational meaning.

8 Conclusion

One of the most important questions in computation is often not

$$\boxed{\text{How do we solve the problem?}}$$

but rather

$$\boxed{\text{Is the problem solvable at all?}}$$

Once existence and uniqueness are established, the actual computation is often considerably simpler. One may still need to solve a linear system, construct an algorithm, perform a factorization, or design an efficient numerical procedure, but the most fundamental question has already been answered: the object we are searching for is known to exist.

The Riesz Representation Theorem carries precisely this burden for linear measurement in a finite-dimensional Euclidean space.

Throughout this note, we have interpreted a linear functional as a mathematical measuring device:

$$\boxed{\text{vector} \xrightarrow{\;f\;} \text{real-valued measurement.}}$$

Thus, if

$$f \in V^*,$$

then $f(x)$ represents a linear scalar measurement of the vector x.

The Riesz Representation Theorem tells us that every such linear measurement can be realized mathematically as an inner-product measurement against one unique vector. That is, for every linear functional f, there exists a unique vector $y \in V$ such that

$$\boxed{f(x) = \langle x, y \rangle \qquad \text{for every } x \in V.}$$

In this sense, the theorem tells us that any linear measurement available in the dual space is mathematically realizable through the geometry of the original vector space.

Whether one can physically construct an instrument to perform that measurement is a separate engineering question. Whether the measurement is cheap, stable, experimentally accessible, or computationally convenient is also a separate question. The theorem addresses the more fundamental mathematical issue:

$$\boxed{\text{the measurement is representable and therefore mathematically possible within the model.}}$$

Once this existence is guaranteed, the remaining task is to determine the representing vector and perform the computation.

Moreover, the value of the measurement cannot depend on the particular coordinate system used to describe the vector. If the basis changes, then the numerical coordinates of both the vector and the functional may change, but the underlying objects remain the same. Consequently,

$$\boxed{f(x) = a_i x^i = a'_i x'^i.}$$

This expresses a fundamental principle of measurement: changing the scale, unit, basis, or coordinate description may alter the numbers used in the calculation, but it must not alter the quantity being measured.

The situation is analogous to measuring the same physical distance in feet or meters. The numerical value changes because the scale changes, but the underlying distance does not. Similarly, in a vector space, a change of basis changes the coordinate representation, while the vector, the functional, and the scalar measurement $f(x)$ remain invariant.

References

[1] Gelfand, I. M., *Lectures on Linear Algebra*, Revised ed., Dover Books on Mathematics, Dover Publications, 1989. ISBN: 978-0-486-66082-0.